

BPHO ROUND 0 · QUESTION PAPER

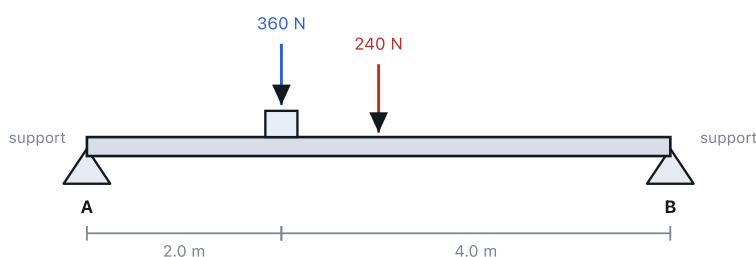
Drill bank section 3

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · A loaded beam on two supports: the reaction at one end **C**

A uniform beam **AB** is 6.0 m long and weighs 240 N. It rests on a support at **A** and a support at **B**. A load of 360 N is placed on the beam 2.0 m from **A**, as shown.



What is the upward force exerted by the support at **A**?

- | | |
|----------------|----------------|
| A 300 N | B 240 N |
| C 360 N | D 480 N |
| E 180 N | |

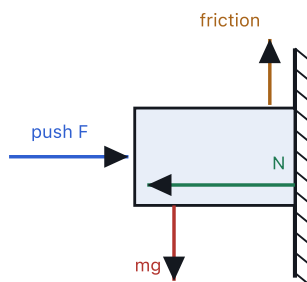
Q2 · Dimensional analysis: the vibration frequency of a liquid droplet **A**

A small droplet of liquid, left to itself, vibrates about its spherical shape. The frequency **f** of that vibration depends only on the surface tension **γ** of the liquid, its density **ρ** , and the radius **r** of the droplet. Which expression is dimensionally a frequency?

- | | |
|-------------------------------------|---------------------------------------|
| A $\sqrt{(\gamma/(\rho r))}$ | B $\sqrt{(\gamma r^3/\rho)}$ |
| C $\sqrt{(\gamma \rho/r^3)}$ | D $\sqrt{(\gamma/(\rho r^3))}$ |
| E $\gamma/(\rho r^3)$ | |

Q3 · Friction against a vertical wall: the smallest push that holds a block up **C**

A block of mass 2.0 kg is held against a rough vertical wall by a horizontal push F , as shown. The coefficient of friction between the block and the wall is 0.40 . The block is on the point of sliding down the wall.

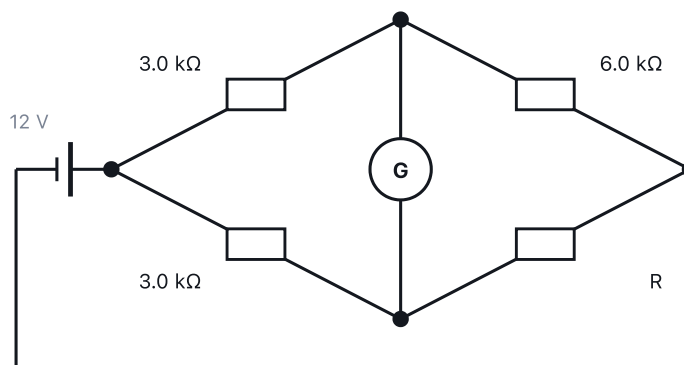


What is the smallest value of F that can hold the block in place?

- | | |
|-------------------------|------------------------|
| A 8.0 N | B 50 N |
| C 20 N | D 80 N |
| E 5.0 N | |
-

Q4 · A balanced bridge: the fourth resistance **H**

The bridge below balances: no current flows through the detector G when R has one particular value.



What is that value of R ?

- | | |
|-------------------------------|-------------------------------|
| A $6.0\text{ k}\Omega$ | B $2.0\text{ k}\Omega$ |
| C $18\text{ k}\Omega$ | D $4.5\text{ k}\Omega$ |
| E $3.0\text{ k}\Omega$ | |
-

Q5 · Estimation: the power needed to climb a 10% hill on a bicycle **A**

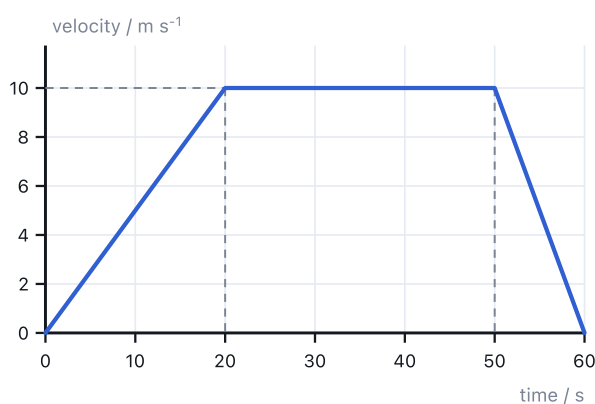
A cyclist and bicycle together have a mass of about 80 kg. The cyclist climbs a hill whose gradient is 10% at a steady 5.0 m/s.

Which value is the closest estimate of the useful power the cyclist is delivering against gravity?

- A** 40 W **B** 2000 W
C 200 W **D** 4000 W
E 400 W
-

Q6 · A three-stage journey: the average speed is a time-weighted mean, not a simple one **B**

A train starts from rest and accelerates uniformly at 0.50 m s^{-2} for 20 s. It then runs at constant speed for 30 s, and finally decelerates uniformly at 1.0 m s^{-2} until it comes to rest. The whole journey is shown below.



What is the average speed for the whole journey?

- A** 7.5 m s^{-1} **B** 10 m s^{-1}
C 8.3 m s^{-1} **D** 9.0 m s^{-1}
E 5.0 m s^{-1}
-

Q7 · Elastic collision: the speed of the heavier target **C**

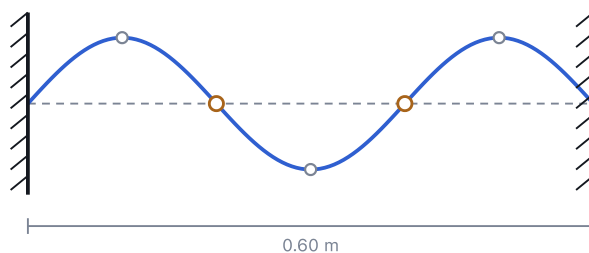
A particle of mass m moves at speed u along a smooth horizontal track. It collides head-on with a stationary particle of mass $3m$. The collision is perfectly elastic.

What is the speed of the particle of mass $3m$ immediately after the collision?

- A** $u/2$ **B** $u/3$
C $u/4$ **D** u
E $3u/4$
-

Q8 · A string fixed at both ends: the third harmonic and its frequency **F**

A string of length 0.60 m is fixed at both ends and vibrates in the pattern shown, which is its third harmonic. The speed of transverse waves on the string is 120 m/s.



What is the frequency of this vibration?

- A** 100 Hz
 - B** 300 Hz
 - C** 200 Hz
 - D** 600 Hz
 - E** 900 Hz
-

Q9 · Apparent depth: how far below the surface a submerged object seems to be **G**

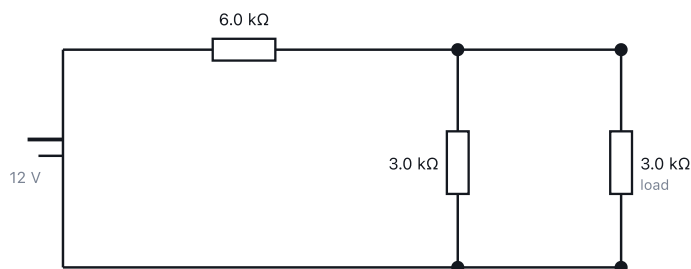
A stone lies on the bottom of a pool of water 2.4 m deep. The refractive index of water is 1.5. An observer looking straight down from directly above sees the stone.

At what depth below the surface does the stone appear to be?

- A** 3.6 m
 - B** 2.4 m
 - C** 0.80 m
 - D** 1.2 m
 - E** 1.6 m
-

Q10 · A loaded potential divider: the voltage actually delivered H

The circuit below shows a 12 V supply connected to a 6.0 k Ω resistor in series with a 3.0 k Ω resistor. A 3.0 k Ω load is connected in parallel with the 3.0 k Ω resistor.



What is the potential difference across the load?

- | | |
|----------------|----------------|
| A 4.0 V | B 4.8 V |
| C 3.0 V | D 2.4 V |
| E 1.2 V | |

Q11 · Ice at -10°C into water at 30°C : whether all the ice can melt, and what is left if it cannot J

0.20 kg of ice at -10°C is added to 0.40 kg of water at 30°C in an insulated container. The specific heat capacity of ice is $2100\text{ J kg}^{-1}\text{ K}^{-1}$, of water is $4200\text{ J kg}^{-1}\text{ K}^{-1}$, and the specific latent heat of fusion of ice is $3.3 \times 10^5\text{ J kg}^{-1}$.

What is the final state of the mixture?

- | | |
|--|--|
| A At 0°C , with 0.060 kg of ice remaining | B At 0°C , with all the ice melted |
| C At 0°C , with 0.14 kg of ice remaining | D At 22°C , with all the ice melted |
| E At 0°C , with none of the ice melted | |

Q12 · Nuclear density: why it is the same in every nucleus K

The radius of a nucleus is given by $r = r_0 A^{1/3}$, where A is the nucleon number and r_0 is a constant.

How does the density of nuclear matter change when A increases from 27 to 216?

- | | |
|--|--|
| A It is unchanged | B It increases by a factor of 8 |
| C It decreases by a factor of 8 | D It increases by a factor of 2 |
| E It decreases by a factor of 2 | |

Q13 · Scaling: by what power of N the total surface area grows when a cube is cut into N cubes **A**

A solid cube is cut up into N identical smaller cubes, where N is a perfect cube.

By what factor does the total surface area of all the pieces together exceed the surface area of the original cube?

A $N^{1/3}$

B $N^{2/3}$

C N

D $N^{1/2}$

E N^3

Q14 · A particle whose deceleration is proportional to its speed: the distance to rest **B**

A particle moves along a straight line with speed v . Throughout its motion its acceleration is $a = -kv$, where k is a positive constant.

The particle has speed u at $t = 0$. How far does it travel before coming to rest?

A uk

B u/k

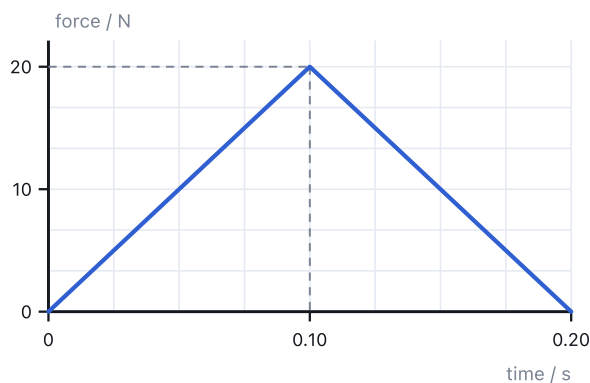
C $u/(2k)$

D $2u/k$

E $u/(k \ln 2)$

Q15 · Impulse from a force-time graph, and the speed it gives a ball **C**

A ball of mass 2.0 kg is at rest on a smooth horizontal surface. A variable force acts on it for 0.20 s ; the graph shows how the force varies with time.



What is the speed of the ball immediately after the force has acted?

A 2.0 m/s

B 0.50 m/s

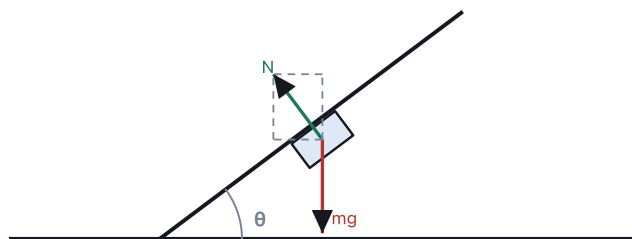
C 4.0 m/s

D 1.0 m/s

E 20 m/s

Q16 · A banked track: the angle at which no friction is needed D

A circular track of radius 30 m is banked at an angle θ to the horizontal, as shown. A car travels round it at 15 m/s without any reliance on friction. Take $g = 10 \text{ m/s}^2$.



cross-section of the track, looking along the road

What is θ ?

- | | |
|---------------------|---------------------|
| A 37° | B 53° |
| C 45° | D 49° |
| E 27° | |

Q17 · Sizing a cable from a breaking stress: the diameter as an algebraic expression E

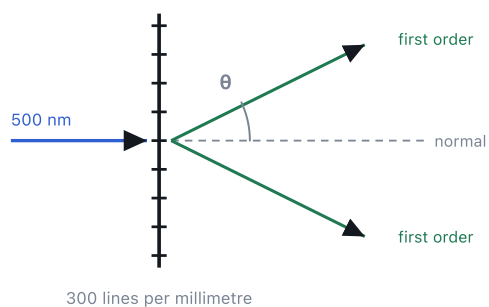
A steel cable is to support a load of mass m . The breaking stress of the steel is σ and the gravitational field strength is g .

Which expression gives the smallest diameter the cable may have?

- | | |
|---------------------------------------|--------------------------------------|
| A $\sqrt{(4mg / (\pi\sigma))}$ | B $\sqrt{(mg / (\pi\sigma))}$ |
| C $4mg / (\pi\sigma)$ | D $\sqrt{(4\pi mg / \sigma)}$ |
| E $\sqrt{(4mg\sigma / \pi)}$ | |

Q18 · The highest diffraction order a grating can produce **F**

A diffraction grating has 300 lines per millimetre. Monochromatic light of wavelength 500 nm falls on it normally, as shown.

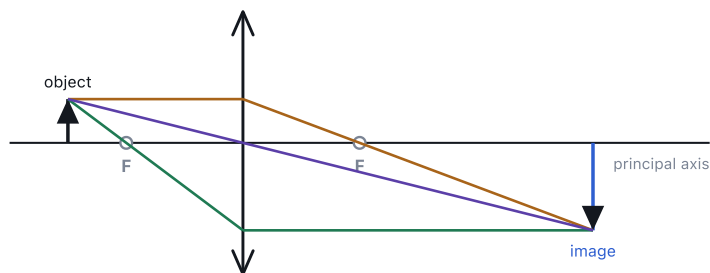


What is the highest order of maximum that can be observed?

- | | |
|-------------|-------------|
| A 7 | B 12 |
| C 13 | D 1 |
| E 6 | |

Q19 · A converging lens: the linear magnification **G**

An object is placed 30 cm from a thin converging lens of focal length 20 cm. The ray diagram shows the construction.



What is the linear magnification produced by the lens?

- | | |
|--------------|---------------|
| A 3 | B 2 |
| C 1.5 | D 0.67 |
| E 0.5 | |

Q20 · Two lamps of different power rating in series: which is brighter H

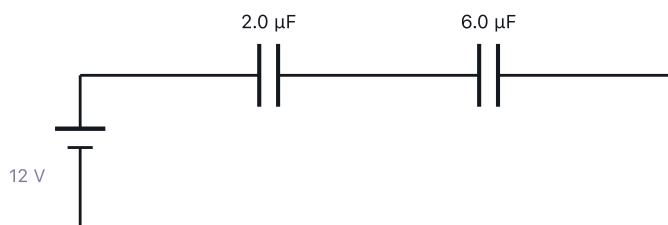
Two filament lamps are marked 60 W and 100 W, both rated for use on 240 V. They are connected in series across a 240 V supply.

Which lamp is the brighter, and why?

- A** The 100 W lamp, because it is rated at the higher power
- B** Neither, because the same current through both makes the powers equal
- C** The 100 W lamp, because its lower resistance lets more current through it
- D** The 60 W lamp, because it has the greater resistance and the current through both is the same
- E** Neither lights, because the two resistances in series are too large for a 240 V supply

Q21 · Capacitors in series: how the supply divides between them I

A 2.0 μF capacitor and a 6.0 μF capacitor are connected in series across a 12 V supply, as shown.



What is the potential difference across the 2.0 μF capacitor?

- A** 9.0 V
- B** 3.0 V
- C** 6.0 V
- D** 4.0 V
- E** 12 V

Q22 · Linear thermal expansion: how much longer a heated rod becomes J

A brass rod is 2.0 m long at 20 °C. The coefficient of linear expansion of brass is 1.9×10^{-5} per °C.

How much longer is the rod at 120 °C?

- A** 7.6 mm
- B** 4.6 mm
- C** 3.8 mm
- D** 1.9 mm
- E** 0.38 mm

Q23 · Mass-energy equivalence: the energy released by a mass defect K

In a nuclear reaction the total mass of the products is less than the total mass of the reactants by $3.6 \times 10^{-30} \text{ kg}$. Take the speed of light as $3.0 \times 10^8 \text{ m/s}$.

How much energy is released?

- | | |
|--|--|
| A $1.1 \times 10^{-21} \text{ J}$ | B $3.2 \times 10^{-13} \text{ J}$ |
| C $4.0 \times 10^{-47} \text{ J}$ | D $2.9 \times 10^4 \text{ J}$ |
| E $3.2 \times 10^{-14} \text{ J}$ | |
-

Q24 · The photoelectric effect: the stopping potential for a given wavelength L

Light of wavelength 400 nm falls on a metal surface whose work function is 2.0 eV.

What is the stopping potential for the emitted photoelectrons?

- | | |
|----------------|-----------------|
| A 3.1 V | B 5.1 V |
| C 2.0 V | D 0.55 V |
| E 1.1 V | |
-

Q25 · Upthrust: what a spring balance reads with the load under water M

A stone has a volume of $2.0 \times 10^{-4} \text{ m}^3$ and a mass of 0.50 kg. It is lowered until it is completely under water and weighed on a spring balance. The density of water is 1000 kg/m^3 and $g = 10 \text{ m/s}^2$.

What does the balance read?

- | | |
|----------------|----------------|
| A 5.0 N | B 2.0 N |
| C 7.0 N | D 3.0 N |
| E 0 N | |